Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

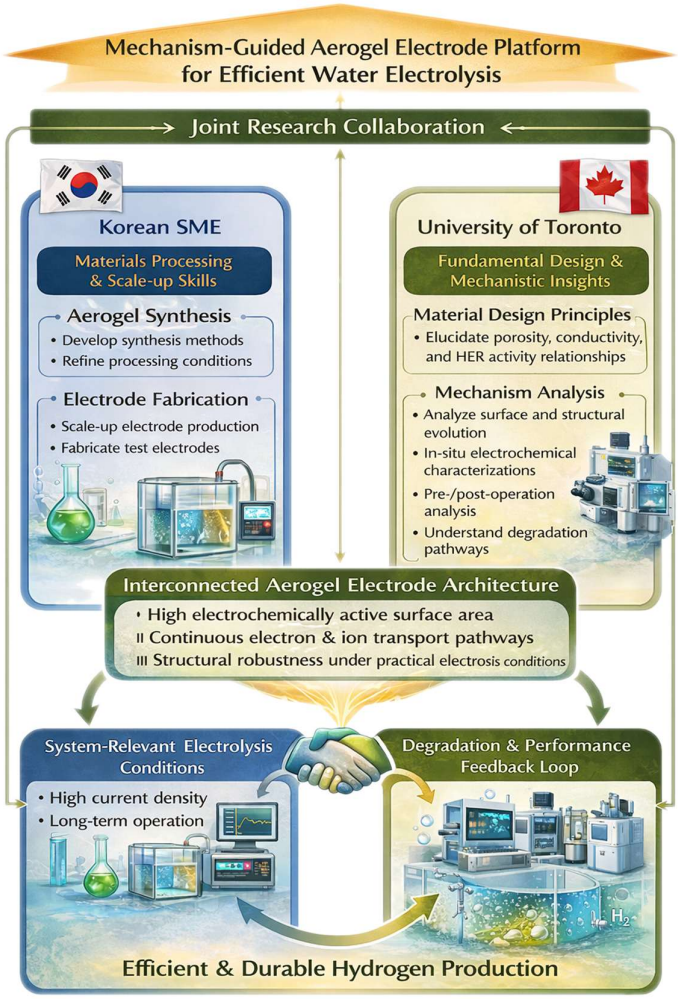
The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Chul B Park, Hani Naguib	Email	park@mie.utoronto.ca
Institute Name	University of Toronto, Department of Mechanical and Industrial Engineering			
Address	5 King's College Road, Toronto, Ontario M5S 3G8			

2. Technology Overview

Title of Proposed Technology	Aerogel-Based Electrocatalyst Platform for Efficient and Durable Water-Splitting Hydrogen Production			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	Energy Materials Technology	New & Renewable Energy	Hydrogen
Concept of Proposed Technology	Despite rapid progress in water electrolysis technologies, Korean SMEs engaged in electrolyzer components and electrode materials continue to face critical on-site challenges, including performance degradation at high current densities, insufficient long-term durability under realistic operating conditions, and limited structural robustness during scale-up from laboratory to pre-pilot systems [1-3]. These issues significantly hinder the reliable and cost-effective implementation of efficient hydrogen production systems. To address these challenges, this proposed technology establishes an aerogel-based electrode material platform for efficient water electrolysis, centered on an interconnected aerogel architecture that provides (i) a high electrochemically active surface area, (ii) continuous electron and ion transport pathways, and (iii) enhanced structural robustness under practical electrolysis operating conditions [4, 5]. By integrating these structural advantages, the proposed platform directly targets the causes of efficiency loss and durability limitations observed in real-world electrolysis environments. Rather than focusing on a single material composition, the proposed technology emphasizes the			

	development of scientifically validated aerogel design principles and performance benchmarks. These guidelines enable industrial partners to rationally tailor material composition, porosity, and interfacial properties in accordance with system-level requirements, including sustained operation at elevated current densities and compatibility with commercial electrolyzer architectures. The resulting aerogel-based electrodes are designed to be electrode-ready and system-compatible, allowing seamless integration into laboratory-scale, prototype, and pre-pilot-scale water electrolysis systems. Through material-to-system validation, this platform supports Korean SMEs in reducing development risks, accelerating technology verification, and bridging the gap between advanced material innovation and practical hydrogen production deployment. Ultimately, the proposed technology contributes to the advancement of efficient, durable, and scalable water electrolysis catalysts, strengthening the industrial competitiveness of Korean SMEs and facilitating the broader implementation of hydrogen production technologies within the national hydrogen ecosystem.
Rationale for Joint R&D Project	 The development of advanced electrode materials for water electrolysis is a critical technological requirement for enabling a sustainable and scalable hydrogen economy. While Korean SMEs possess strong capabilities in materials processing, component manufacturing, and scale-up, the

	successful translation of emerging aerogel materials into electrode-ready technologies for hydrogen production systems requires a level of mechanistic understanding that extends beyond conventional industrial R&D. In particular, aerogel-based electrodes experience dynamic surface reconstruction, phase evolution, and microstructural changes under realistic electrolysis operating conditions such as high current densities, prolonged operation, and fluctuating load environments, which strongly influence catalytic activity, durability, and long-term system compatibility. Without a clear understanding of these processes, material optimization and system integration remain largely empirical, increasing development time and technical risk for SMEs. Addressing these challenges necessitates advanced characterization, operando/ex-situ mechanistic analysis, and structure–property–performance correlation, which are typically beyond the scope of in-house SME research infrastructure. In this context, an international collaboration that combines the industrial scale-up expertise of Korean SMEs with the University of Toronto's strength in mechanistic electrochemical analysis provides a uniquely effective framework for overcoming these limitations. Through this joint R&D project, the University of Toronto will provide mechanism-driven design guidelines, performance validation under system-relevant conditions, and scientifically grounded optimization strategies for aerogel-based electrode materials. This collaboration will reduce technical uncertainty and development risk for Korean SMEs, enabling them to accelerate material-to-system translation. Ultimately, the joint project establishes a pathway to develop aerogel electrode technologies that are not only high-performing at the laboratory scale but also robust, scalable, and compatible with practical water electrolysis platforms, thereby strengthening the competitiveness of Korean SMEs within the global hydrogen technology ecosystem.
Key Words	1. Aerogel electrocatalyst 2. Water electrolysis 3. Hydrogen production 4. System-compatible electrode materials 5. Electrode durability

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	The objective of this joint R&D project is to develop system-relevant, aerogel-based electrode materials for water electrolysis by establishing scientifically validated material design principles and performance criteria that can be directly adopted by Korean SMEs for practical hydrogen production systems. Specifically, the project aims to design and optimize interconnected aerogel architectures with controlled porosity, continuous electron and ion transport pathways, and tailored electrochemical			

	properties, enabling high catalytic activity, efficient charge transport, and sustained electrode durability under realistic water electrolysis operating conditions, including high current densities and long-term operation. In parallel, the project seeks to establish quantitative structure–performance–durability relationships through mechanism-driven investigation of surface reconstruction, interfacial reactions, and degradation pathways occurring during electrolysis. These mechanistic insights will be systematically translated into clear material design guidelines and performance benchmarks for the development of electrode-ready aerogel materials by Korean SMEs. Ultimately, this R&D project aims to bridge fundamental material innovation and practical hydrogen production technologies by enabling the integration and validation of advanced aerogel-based electrodes in laboratory-scale and pre-pilot-scale water electrolysis systems, thereby accelerating material-to-system translation while reducing technical uncertainty and development risk for industrial partners.
R&D Target Specifications	1. Material structural properties ✓ Aerogel porosity of **≥90%**, ensuring a highly interconnected three-dimensional porous network. ✓ Specific surface area of **≥100 m^2 g^{-1}**, as measured by BET analysis, suitable for high-density electrochemical active sites. 2. Electrochemical Activity ✓ Achieve stable operation of aerogel-based electrodes at a current density of **≥50 mA cm^{-2}** under alkaline water electrolysis conditions. ✓ Attain an overpotential of **≤150 mV at 50 mA cm^{-2}** for hydrogen evolution reactions (HER) in electrode-level testing. 3. Reaction Kinetics and Charge Transfer ✓ Demonstrate a Tafel slope **≤70 mV dec^{-1}**, indicating favorable reaction kinetics. ✓ Reduce **charge transfer resistance (R_{ct}) by ≥30%** compared to benchmark powder-type electrode materials 4. Durability and Structural Stability ✓ Maintain **≥90% of the initial current density after 100 h** of continuous electrolysis operation at system-relevant conditions ✓ Preserve the aerogel porous structure without significant collapse or delamination after long-term testing, as confirmed by post-operation structural analysis. 5. System Compatibility ✓ Demonstrate electrode functionality in **laboratory-scale single-cell electrolysis setups** operating at current densities **≥100 mA cm^{-2}**. ✓ Verify compatibility with practical electrode fabrication processes suitable for pre-pilot-scale evaluation by Korean SME partner.

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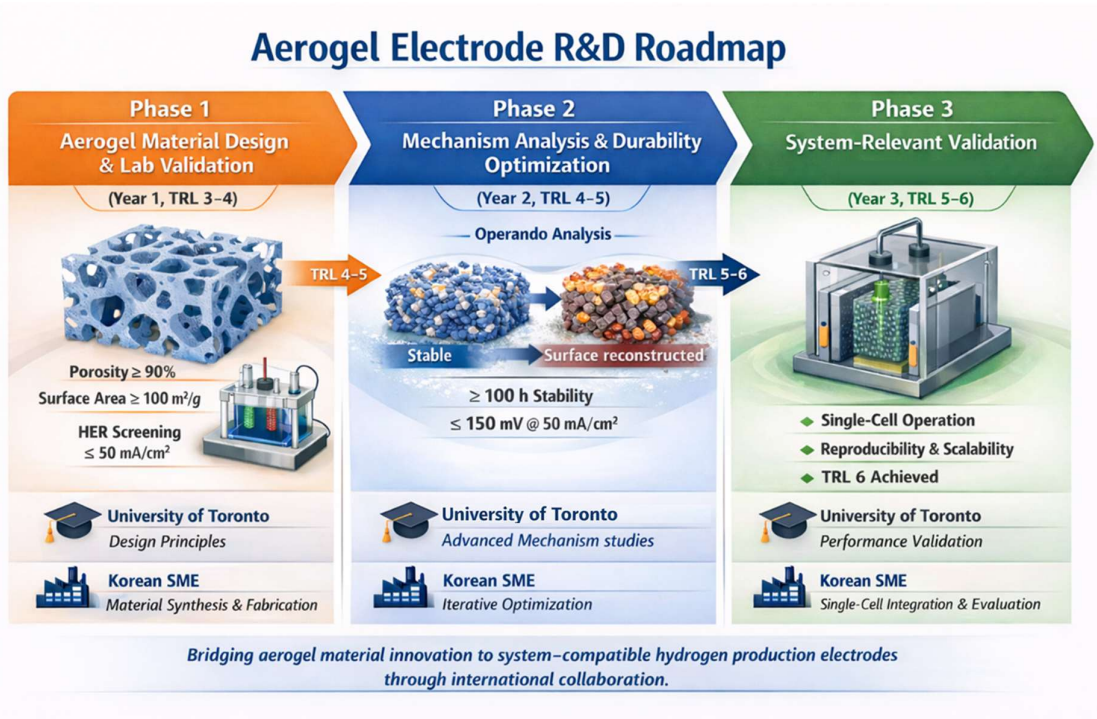
R&D Plan



[Phase 1 (Year 1). Aerogel material design and laboratory-scale validation]

Research approach & Key tasks

- Design transition metal (3*d* series) oxide aerogel material with controlled porosity, surface area, composition, and conductivity for water electrolysis electrodes.
- Synthesis and processing of aerogel materials suitable for electrode fabrication and laboratory scale evaluation.
- Perform screening of electrochemical evaluations to confirm baseline material performance for HER relevant to water electrolysis electrodes.

Roles and collaboration

University of Toronto:

- Establish aerogel-based material design principles linking porous structure to electrochemical performance.
- Conduct fundamental electrochemical characterization and identify key structure–property relationships.

Korean SME:

- Develop reproducible aerogel synthesis routes and assess process feasibility.
- Fabricate electrode-ready aerogel materials based on proposed design criteria.

Methodologies

- Material structural characterization (BET, porosity analysis)
- Electrode-level electrochemical testing (HER activity, Tafel, and EIS analysis)

Milestones

- Aerogel materials with porosity **≥90%** and surface area **≥100 m^2 g^{-1}**.
- Baseline HER performance established at current densities up to **50 mA cm^{-2}**.

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[Phase 2 (Year 2). Mechanism analysis and durability-oriented optimization]

Research approach & Key tasks

- Optimization of aerogel composition and structure for improved HER kinetics and durability under continuous electrolysis operation.
- Investigation of reaction-induced structural and surface evolution under electrolysis conditions to identify the surface reconstruction relevant to long-term operations.

Roles and collaboration

University of Toronto:

- Perform advanced mechanistic studies to elucidate surface reconstruction and degradation pathways.
- Provide durability-oriented material design guidelines based on mechanism-driven insights.

Korean SME:

- Refine aerogel material processing and electrode fabrication following provided guidelines.
- Conduct extended durability testing and feedback results for iterative optimization.

Methodologies

- Long-term electrochemical testing (≥100 h operation).
- Pre- and post-operation structural and electrochemical analysis.

Milestones

- Achieve HER overpotential **≤150 mV at 50 mA cm^{-2}**.
- Baseline Demonstrate **≥90%** performance retention after **100 h operation** under electrolysis conditions relevant to practical systems.

[Phase 3 (Year 3). System-relevant validation and TRL advancement]

Research approach & Key tasks

- Validation of optimized aerogel electrodes under system-relevant electrolysis conditions, including single cell systems.
- Assessment of material consistency, reproducibility, and system compatibility with practical electrolysis.

Roles and collaboration

University of Toronto:

- Validate electrochemical performance under system-relevant conditions and provide final performance benchmarks.
- Support interpretation of results for TRL assessments.

Korean SME:

- Integrate aerogel materials into pre-pilot -scale single-cell electrolysis systems.
- Evaluate scalability, reproducibility, and applicability for pre-pilot-scale deployment.

Methodologies

- Single-cell electrolysis testing.
- Performance benchmarking and reproducibility assessment.

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	Milestones ● Stable operation of aerogel-based electrodes in single-cell electrolysis systems. ● Advancement to TRL 6, demonstrating system-compatible aerogel electrode materials validated in a relevant electrolysis environment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The University of Toronto will serve as the research institute responsible for providing fundamental scientific and technical leadership for this joint R&D project. Its primary roles include establishing material design principles for aerogel-based electrode materials, conducting advanced electrochemical and structural analyses, and elucidating reaction and degradation mechanisms under water electrolysis operating conditions. The research institute will lead performance validation, durability assessment, and mechanism-driven interpretation to reduce technical risks associated with aerogel material development. In addition, the University of Toronto will provide standardized evaluation protocols, analytical infrastructure, and technical guidance to ensure that the aerogel materials developed by the Korean SME meet system-relevant performance and durability requirements for practical water electrolysis applications.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, processing, and scale-up of aerogel-based materials in accordance with the design guidelines established through the joint R&D collaboration. Its roles include laboratory- and pilot-scale material synthesis, electrode fabrication, and reproducibility assessment, with an emphasis on manufacturability and process consistency. The SME will conduct practical performance testing and durability evaluation in system-relevant electrode and cell configurations, while assessing scalability, manufacturability, and commercialization feasibility. The Korean SME will actively collaborate with the University of Toronto by incorporating feedback from mechanistic studies and performance validation, thereby accelerating the translation of aerogel material innovations into electrode-ready and industry-relevant products for water electrolysis applications.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to generate significant technological and industrial outcomes by enabling the development of system-compatible aerogel-based electrode materials for water electrolysis. From a technological perspective, the project will establish scientifically validated design principles for aerogel materials with controlled porosity, surface area, and electrochemical durability, supporting stable hydrogen production under practical operating conditions. The integration of mechanism-driven analysis and system-relevant validation will reduce technical uncertainty and accelerate the transition of aerogel materials from laboratory-scale concepts to electrode-ready products. From a commercialization perspective, the project will provide Korean SMEs with a differentiated material platform that can be supplied to water electrolysis system manufacturers as high-performance electrode materials. By demonstrating reproducible performance, durability, and compatibility with practical electrode fabrication processes, the developed aerogel materials are expected to enhance the competitiveness of domestic SMEs in the hydrogen value chain and create opportunities for downstream partnerships and technology licensing. In terms of broader industrial and economic impact, this international joint R&D project will strengthen Korea's technological foundation in hydrogen production by fostering collaboration between global research institutions and domestic material companies. The outcomes will contribute to the localization and advancement of core electrode materials, support the growth of hydrogen-related industries, and align with national strategies for carbon neutrality and sustainable

	energy transition. Furthermore, the project will serve as a model for effective international cooperation that links fundamental material science with industrial innovation, delivering long-term value to Korea's hydrogen industry eco-system.

6. References

(1) S.-M. Jeon, "'진짜 수소'는 따로 있다?" 전기로 물을 분해하는 '수전해' 기술 총정리, SK ecoplant, 2024.07.03, https://news.skecoplant.com/board/16794.

(2) S. Xu, P. Zhang, L. Li, M.-W. Moon, C.-H. Chung, H. Li, J. Y. Lee, P. J. Yoo, Challenges and Emerging Trends in Hydrogen Energy Industrialization: From Hydrogen Evolution Reaction to Storage, Transportation, and Utilization. Small 2025, 21, 2502000. https://doi.org/10.1002/smll.202502000.

(3) B. Gim, J. Kim, H. Ko, Economic Evaluation of Domestic Low-Temperature Water Electrolysis Hydrogen Production. Trans. Of the Korean Hydrogen and New Energy Society, 2011, 22, 4, 559-567, https://doi.org/10.7316/khnes.2011.22.4.559.

(4) Kim, T., Jung, H., Choi, H., Lee, W., Patil, U.M., Parale, V.G., Kim, Y., Kim, J., Kim, S.-H., &Park, H.-H., Partially oxidized inter-doped RuNi alloy aerogel for the hydrogen evolution reaction in both alkaline and acidic media. Materials Horizons, 2024. 11(17): p. 4123. https://doi.org/10.1039/D4MH00242C.

(5) S. Yan, X. Chen, W. Li, M. Zhong, J. Xu, M. Xu, C. Wang, N. Pinna, X. Lu, Highly Active and Stable Alkaline Hydrogen Evolution Electrocatalyst Based on Ir-Incorporated Partially Oxidized Ru Aerogel under Industrial-Level Current Density. Adv. Sci. 2024, 11, 2307061. https://doi.org/10.1002/advs.202307061.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization